

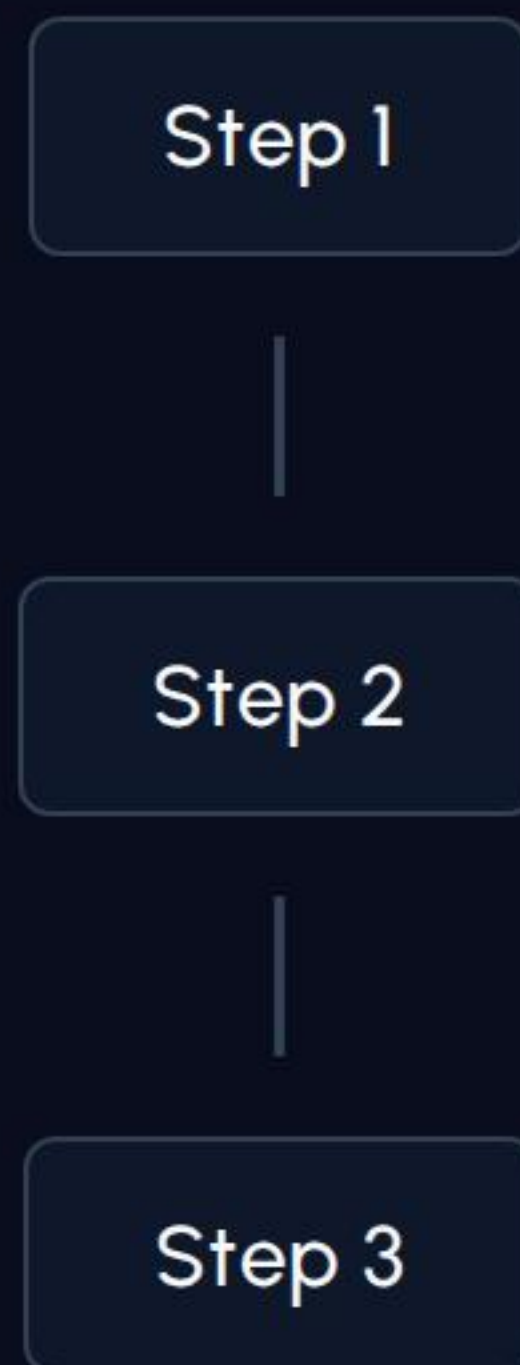
Logic in Pictures

A Visual Guide to Conditionals

PROGRAM FLOW CONTROL

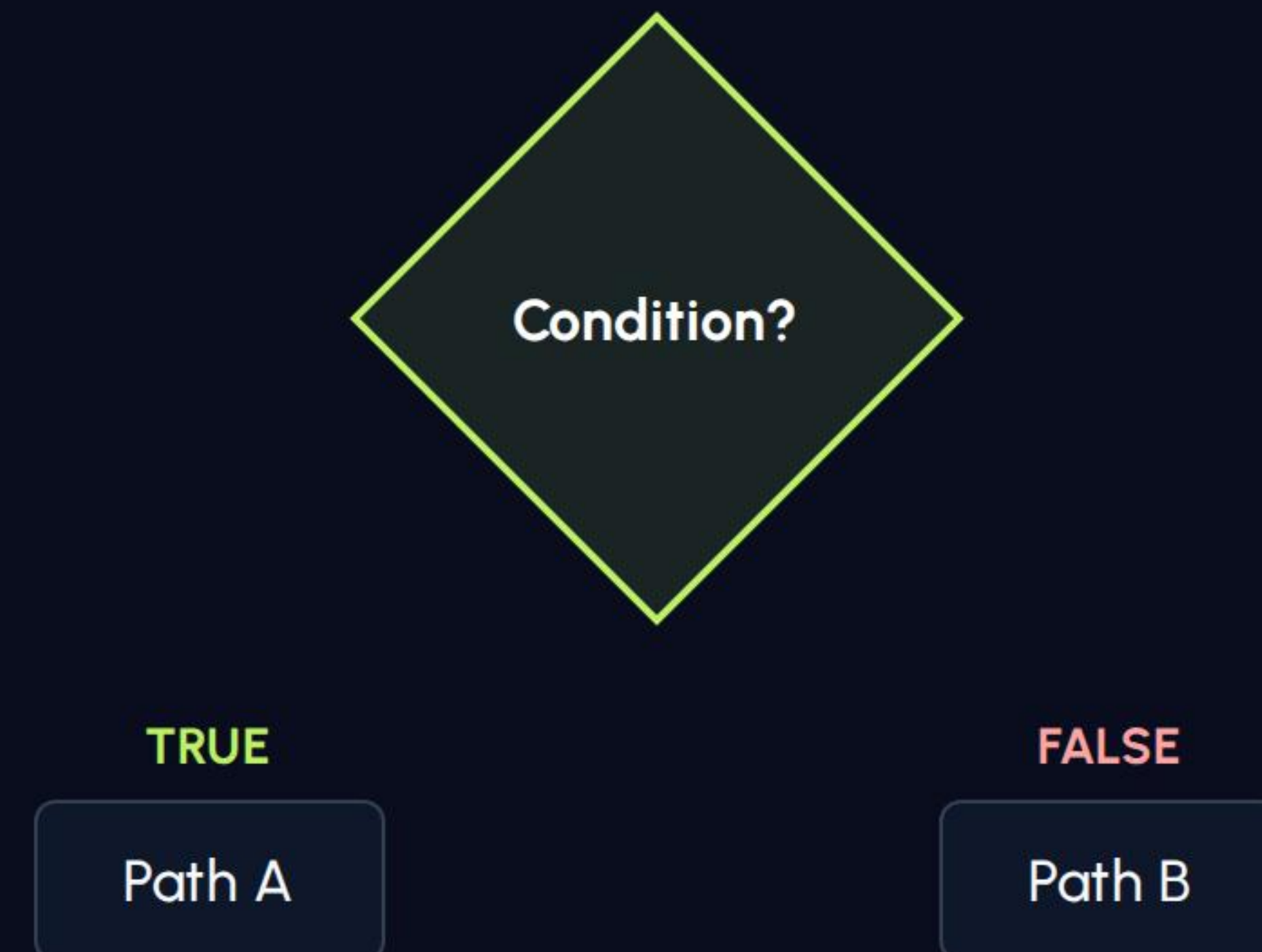
Linear Execution

Step-by-step sequence.



Decision Making

Logic-based redirection.



| THE "IF" STATEMENT

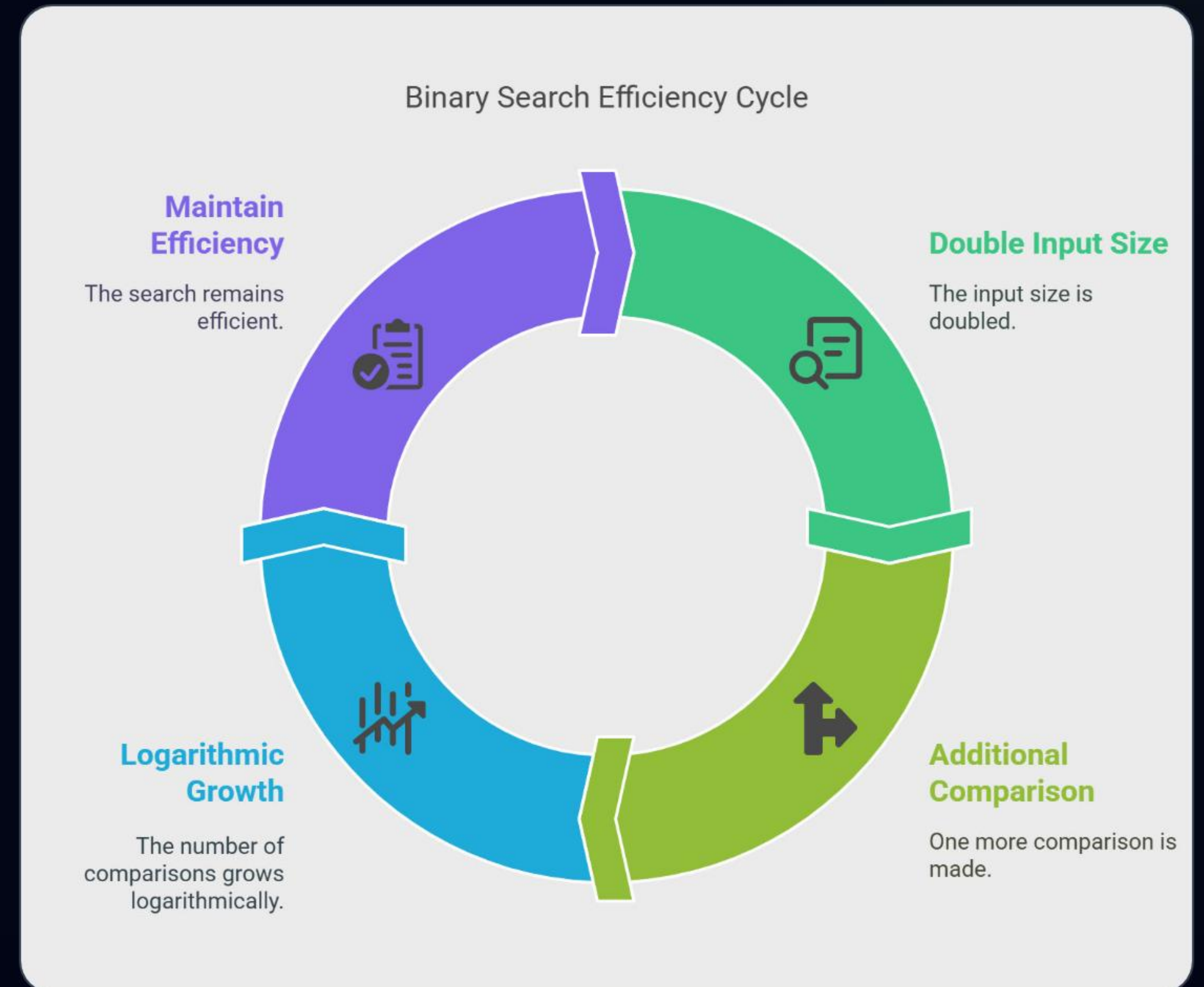


The code inside the `if` runs only if the condition evaluates to `True`.

```
if temperature < 20:  
    print("Cold")
```

```
print("Proceeding...")
```

| THE "IF-ELSE" STATEMENT



| MULTIPLE PATH CHAINS



Only the **first True block** executes. The program skips the rest of the chain instantly.

| COMPARISON REFERENCE

Operation	Syntax	Result Example
Equality	<code>=</code>	<code>5 = 5</code> is True
Inequality	<code>≠</code>	<code>5 ≠ 10</code> is True
Greater Than	<code>></code>	<code>10 > 2</code> is True
Inclusion	<code>in</code>	<code>"a" in "apple"</code> is True

| LOGIC COMBINATIONS



AND

Strictest check. Requires **every** condition to be True.



OR

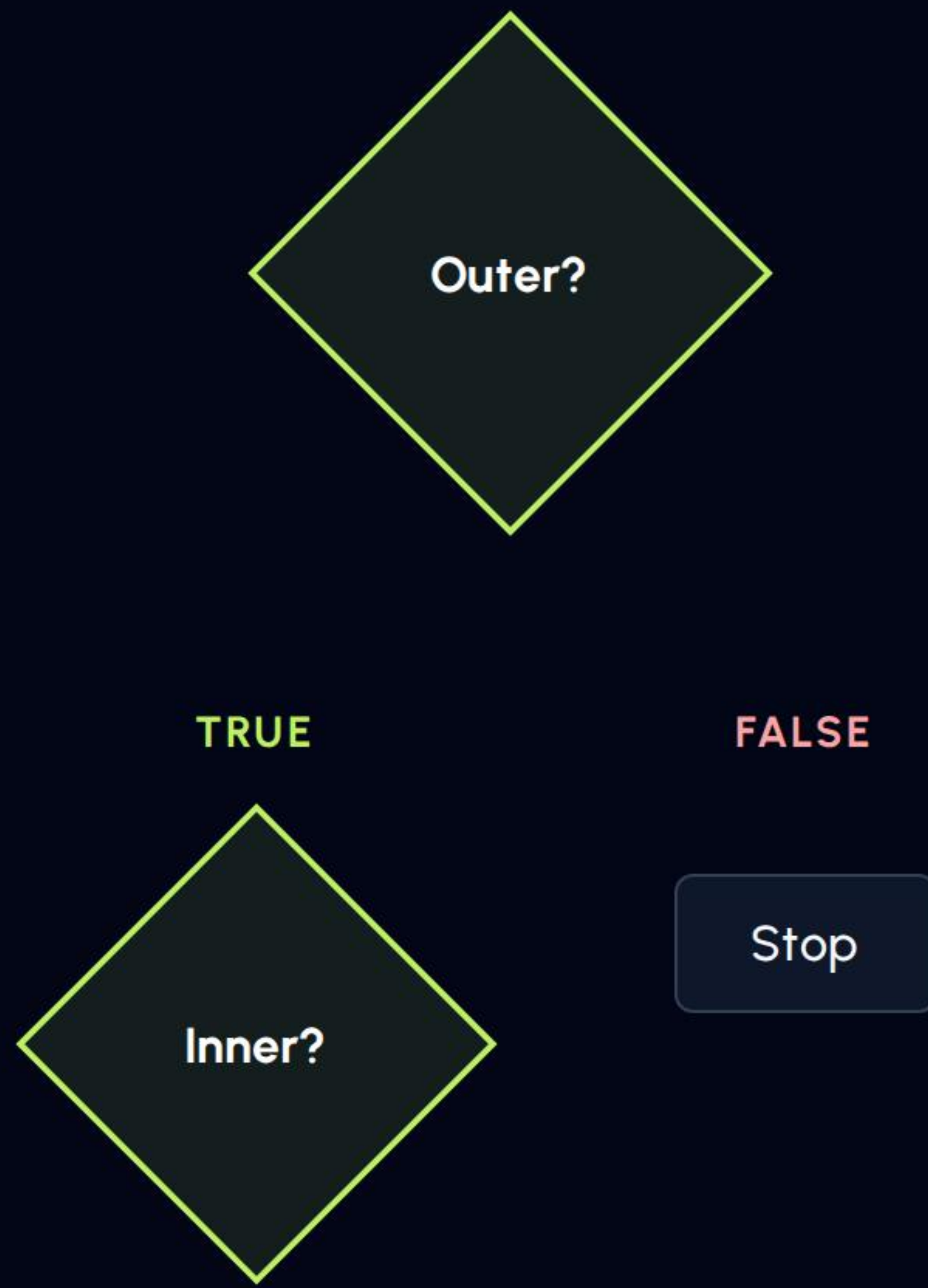
Flexible check. Requires **any** condition to be True.



NOT

Inverse check. Reverses **True** to **False** and vice-versa.

| LAYERED LOGIC (NESTED)



Conditional Hierarchy

The **Inner** condition is only reached if the **Outer** condition evaluates to **True**.

```
if age ≥ 18:
    if has_id:
        print("Welcome")
    else:
        print("No ID")
```

| GUARD CLAUSE PATTERN

The Arrow Pattern (Bad)

Heavy nesting makes logic hard to follow.

```
if user_active:
    if data_exists:
        if server_up:
            process()
```

The Guard Pattern (Good)

Exit early to keep main logic readable.

```
if not active: return
if not data: return
if not server: return

process()
```

| TERNARY EXPRESSION

```
val_if_true if condition else val_if_false
```

```
msg = "Pass" if score ≥ 50 else "Fail"
```

| SYNTAX PITFALLS

= vs ==

`=` assigns a value.
`==` checks equality.

Missing :

The colon must follow every `if`,
`elif`, and `else`.

Indentation

Python defines blocks by spacing,
not braces `{}`.

TRACE VISUALIZATION

Input: 5

1. `number > 0?` → **TRUE**

→ `print("Positive")`

2. Skip all `elif` and `else`.

Input: 0

1. `number > 0?` → **FALSE**

2. `number < 0?` → **FALSE**

3. Run `else` → **TRUE**

→ `print("Zero")`

| IMAGE SOURCES



https://ik.imagekit.io/upgradl/abroad-images/imageCompo/images/__visual_selection_2025_06_24T125046_90908RTGA.png?pr=true

Source: www.upgrad.com